

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-309-1

28-02-2023

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SPECIFICATION

FOR

PERSONAL PROTECTIVE EQUIPMENT

HAND PROTECTION GLOVES

(ELECTRICAL INSULATING / COMPOSITE)

Issue: Feb.-2023/ Rev- 1

- This revision contains option items that must be selected by EDC before bidding.

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EEHC DISTRIBUTION MATERIALS SPECIFICATION**EDMS 30-309-1****28-02-2023****1. SCOPE**

This specification covers the minimum technical requirements for design, manufacturing, testing, and supply of electrical insulating gloves as personal protective equipment (PPE). The required gloves are intended to protect the wearer's hand against electrical shock during live work in the Egyptian Electricity Distribution Companies (EDCs). This specification also covers electrical insulating gloves with additional integrated mechanical protection (composite gloves).

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
IEC 60903	Live working – Electrical insulating gloves
IEC 60060-1	High voltage test techniques – part 1 General definitions and test requirements
IEC 60060-2	High voltage test techniques – part 2 Measuring techniques
ISO 23529	Rubber- general procedures for preparing and conditioning test pieces for physical test methods
IEC 60212	Standard conditions for use prior to and during the testing of solid electrical insulating materials
IEC 60417	Graphical symbols for use on equipment
IEC 61318	Live working - Methods for assessment of defects and verification of performance applicable to tools, devices and equipment
IEC 61477	Live working - Minimum requirements for the utilization of tools, devices and equipment

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The safety behavior of the required coverall shall be guaranteed under the environmental conditions given in Table (2).

Table (2)

Minimum ambient temperature	-5 °C
Maximum ambient temperature	+50 °C
Maximum relative humidity	95 %
Storage condition	Dry place
Working environment	Humid tropical climate with heavily polluted atmosphere

4. DEFINITIONS and CLASSIFICATIONS

- Electrical insulating glove is a glove used for the protection of the worker against electric shock.
- The maximum use voltage is the a.c. voltage (r.m.s) rating of the insulating glove that designates the maximum nominal voltage of the energized system that may be safely worked. The nominal voltage is equal to the phase-to-phase voltage on three-phase circuits or phase-to-ground voltage on single-phase circuit.
- Composite glove is an electrical insulating glove made with additional integrated mechanical protection of the glove against punctures, tears, abrasions ... etc.

4.1 Class of Insulating Gloves

- Class of the insulating glove shall be designated according Table (3). The maximum thickness on the flat surface of a glove (no ripped area if present) shall be as given in Table (3) in order to obtain the appropriate flexibility.

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Table (3)

Class	AC V rms	DC V	Maximum thickness (mm)		Color code
			Insulating glove	Composite glove	
00	500	750	0.50	1.8	Beige
0	1000	1500	1.00	2.3	Red
1	7500	11250	1.50	2.8	White
2	17000	25500	2.30	3.3	Yellow
3	26500	39750	2.90	3.6	Green
4	36000	54000	3.60	4.2	Orange

4.2 Category of Insulating Gloves

- Insulating gloves shall be marked according to acid, oil, ozone and extreme low temperature as mentioned in Table (4).

Table (4)

Category	Resistant to
A	Acid
H	Oil
Z	Ozone
R	Acid, Oil, Ozone
C	Extremely low temperature
F	Leakage current (for long gloves only)

4.3 Standard Lengths of Insulating Gloves

- Length of the insulating gloves shall be manufactured according Table (5).

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Table (5)

Class	Standard Length * (mm)				Length of long gloves ** (mm)
00	280	360	-	-	800
0	280	360	410	460	800
1	-	360	410	460	800
2	-	360	410	460	800
3	-	360	410	460	800
4	-	-	410	460	800

* The permissible variation in length shall be ± 15 mm for any class** For long gloves the tolerance in length shall be ± 20 mm.

4.3 Standard Sizes of the Gloves

- Sizes of the insulating gloves shall be manufactured according Table (6).

Table (6)

Glove size	Palm Circumference * (mm)
7	178
7.5	191
8	203
8.5	216
9	229
9.5	241
10	254
10.5	267
11	279
11.5	292
12	305

* The tolerance in size shall be ± 13 mm.

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5. DESIGN REQUIREMENTS

- Gloves are made of elastomer. Elastomer is macromolecular material which returns rapidly to its initial dimensions and shape after substantial deformation by a weak stress and release of the stress.
- Glove design shall be such that the fingers are slightly bent in a position corresponding to the position the hand forms when while holding an object.
- An electrical insulating gloves shall have the electrical, mechanical, chemical, thermal, ageing and flame retardancy characteristics conforming to the relevant standard specified.
- The gloves shall be manufactured with a cuff.
- According to __ EDC requirements, the Cuff design can be contour (refer to Figure 1) or straight (refer to Figure 2) for normal use as shown below. Also, gloves can be requested with cuff roll as indicated in Figure 2.
- Contour gloves shall be designed in such a way as to facilitate the bending of the arm and shall be intended for gloves that extend to or above the elbow. The difference between the maximum and minimum lengths (dimension C in Figure 1) shall be $50 \text{ mm} \pm 6 \text{ mm}$ except for long gloves where this difference shall be $100 \text{ mm} \pm 12 \text{ mm}$.

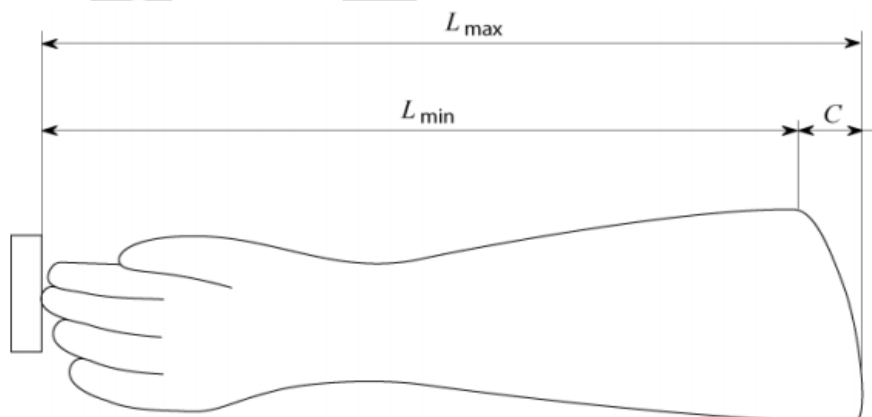


Figure (1) Contour cuff design

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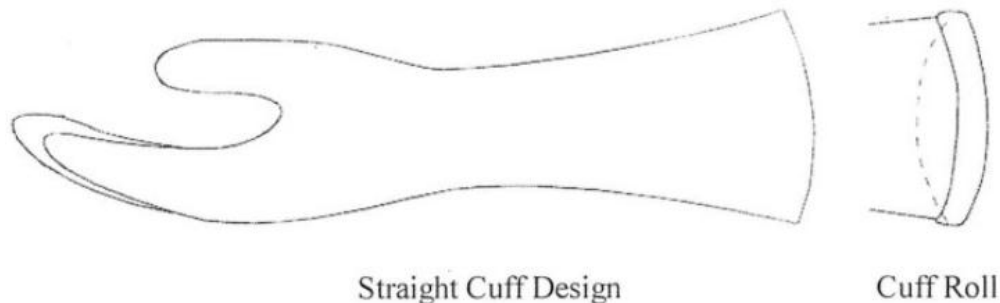


Figure (2)

- The gloves shall be produced by a seamless process and shall be free on both inner and outer surfaces from harmful physical irregularities such as pinholes, cracks, blisters, cuts, conductive embedded foreign matter, creases, pinch marks, voids (entrapped air), prominent ripples, and prominent mould marks that can be detected by thorough test and inspection.
- Palm and finger surfaces designed to improve the grip shall not be considered as irregularities.
- The glove should be recently fabricated. Manufacturing date should not exceed two years before the bid closing date.

6. MARKING

- Each glove shall bear a label and/or marking giving the following information:
 - Symbol IEC 60417-5216 – suitable for live working (double triangle as shown in Figure 3).



Figure (3) Symbol IEC 60417-5216

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- Number of relevant IEC standard with year of publication (four digits); (for example IEC 60903:2014)
 - Name, trademark or identification of the manufacturer.
 - Category
 - Size
 - Class
 - Month and year of manufacture
 - Composite gloves shall also identified with mechanical symbol (hammer)
- In addition, each glove shall provide an area permitting the marking of the date of current inspection or date of next required inspection and test, or any other suitable means to identify the date the glove put into service and the dates of periodic inspection and test.
 - The marking or label shall not impair the quality of the glove; it shall be indelible and shall remain visible after being subjected to a durability test as specified in the standard. All such marking shall be confined to the cuff portion of the glove and shall be non-conducting.
 - When the color code for markings/symbols is used, it shall correspond to the given in Table (3).
 - Marking shall be clearly visible and legible without additional magnification.

7. TESTING and INSPECTION

- Gloves shall be subjected to the tests described in Table (7) according to the recent relevant standard:

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Table (7)

S. No.	Test		Reference Clause in IEC 60903
1	Visual and dimensional inspection	Classification	5.2.2
		Dimensions	5.2.3
		Thickness	5.2.4
		Workmanship and finish	5.2.5
2	Marking	Visual and dimensional inspection	5.3.1
		Durability of marking	5.3.2
3	Packaging and instructions for use		5.4
4	Mechanical tests (Type tests)	Tensile strength and elongation at break	5.5.2
		Resistance to mechanical puncture.	5.5.3
		Tension set.	5.5.4
5	Dielectric tests	AC proof (Type test)	5.6.1
		AC withstand (Type test)	5.6.1
		Routine dielectric test	5.6.2
6	Ageing test		5.7
7	Thermal tests	Low temperature tests	5.8.1
		Flam retardancy test	5.8.2
8	Tests on gloves with special properties	Category A – Acid resistance	5.9.1
		Category H – Oil resistance	5.9.2
		Category Z – Ozone resistance	5.9.3
		Category C – Extremely low temperature resistance	5.9.4
		Category F – Leakage current resistance	5.9.5
9	Specific mechanical testing for composite gloves	Abrasion resistance	5.10.1
		Cutting resistance	5.10.2
		Tear resistance	5.10.3

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- The supplier shall guarantee the supplied materials against any defects arising out of faulty design or workmanship, or of defective material for a period of one (1) year at least from the date of delivery.

9. SUBMITTALS

- The supplier shall complete and return one copy of the attached Data Schedule.
- Technical data and original catalogues shall be submitted with the offer.
- **One (pair) specimen** of the offered glove shall be submitted with the offer for evaluation prior to issuance of the purchase order. The specimen should correspond in all respects with the coverall they intend to supply. Color, Size and dimension verification along with indelible markings and finishing shall be checked.
- A valid approval certificate issued by Egyptian Electricity Holding Company.
- Recent test reports should be submitted with the offer.

10. PACKING and DELIVERY

- Each pair of gloves shall be packaged in an individual package of sufficient strength to properly protect the product from damage.
- The type of packaging suitable for transport shall be defined by the manufacturer. The outside of the container or package shall be marked with the name of manufacturer or supplier, the classification, category, size, length and cuff design.
- The supplier should provide storage and use instructions in Arabic.

11. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

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TECHNICAL DATA SCHEDULE
PERSONAL PROTECTIVE EQUIPEMENT (PPE)
(HAND PROTECTION GLOVES)

__ EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	__ EDC Requirement*	Vendor Offer**
1	Manufacturer's name	Specify	
2	Model	Specify	
3	Country of Origin	Specify	
4	Applicable Standard	IEC 60903	
5	Glove type	Electrical insulating / Composite	
6	Class	As listed in Table 3	
7	Category	As listed in Table 4	
8	Glove design	Straight / Contour	
9	Cuff roll	With / without	
10	Material	Specify	
11	Glove size	As listed in Table 6	
12	Glove length (mm)	As listed in Table 5	
13	Max. thickness (mm)	As listed in Table 3	

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14	Color code	As listed in Table 3	
15	Year of Manufacture	Manufactured within two years before the bid closing date.	
16	Marking / label	Comply with clause 6?	
17	Test Reports	Comply with clause 7?	
18	Specimen	Comply with clause 9?	
19	Guarantee	One year at least from delivery	
20	Packing and delivery	Comply with clause 10?	
21	Submittals	Comply with clause 9?	

(*) - Values to be provided / proposed by the vendor.

(**) – Please, provide explanations for deviations if any. (Use separate sheet if necessary.)

I / We guarantee the technical data given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		